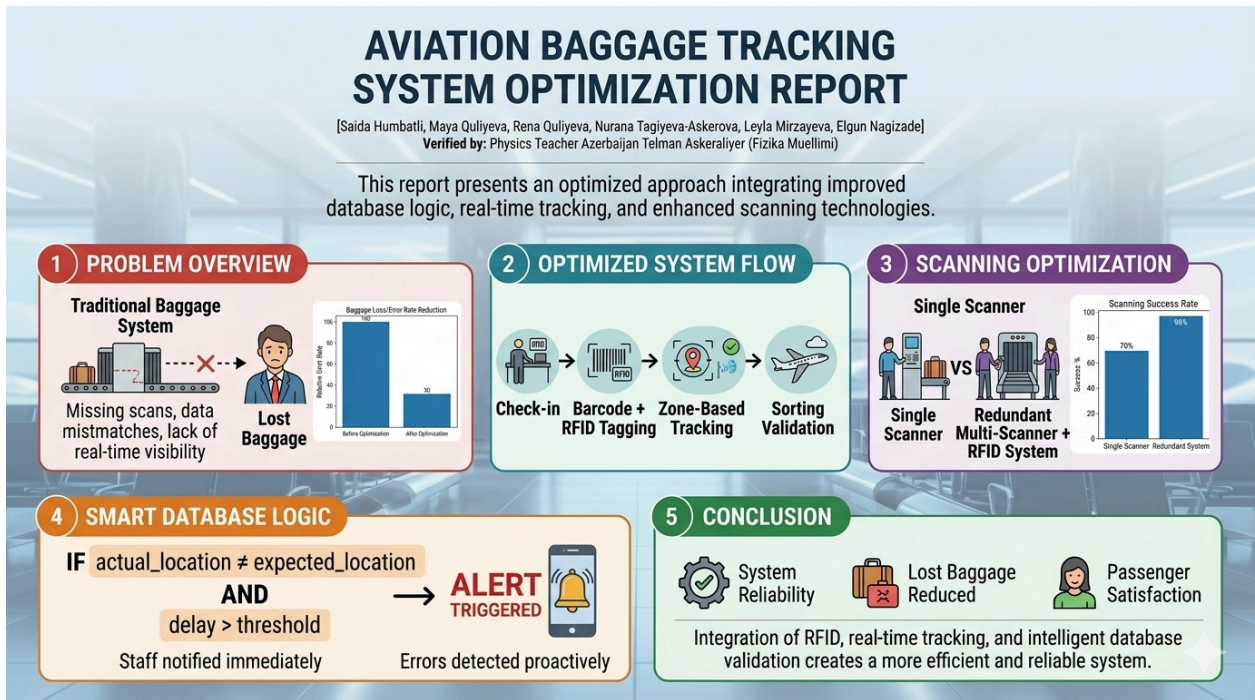


Baggage Tracking System Optimization Report

Topic description: Aviation system baggage tracking optimization| Saida Humbatli, Maya Quliyeva, Rena Quliyeva, Nurana Tagiyeva-Askerova, Leyla Mirzayeva, Elgun Nagizade
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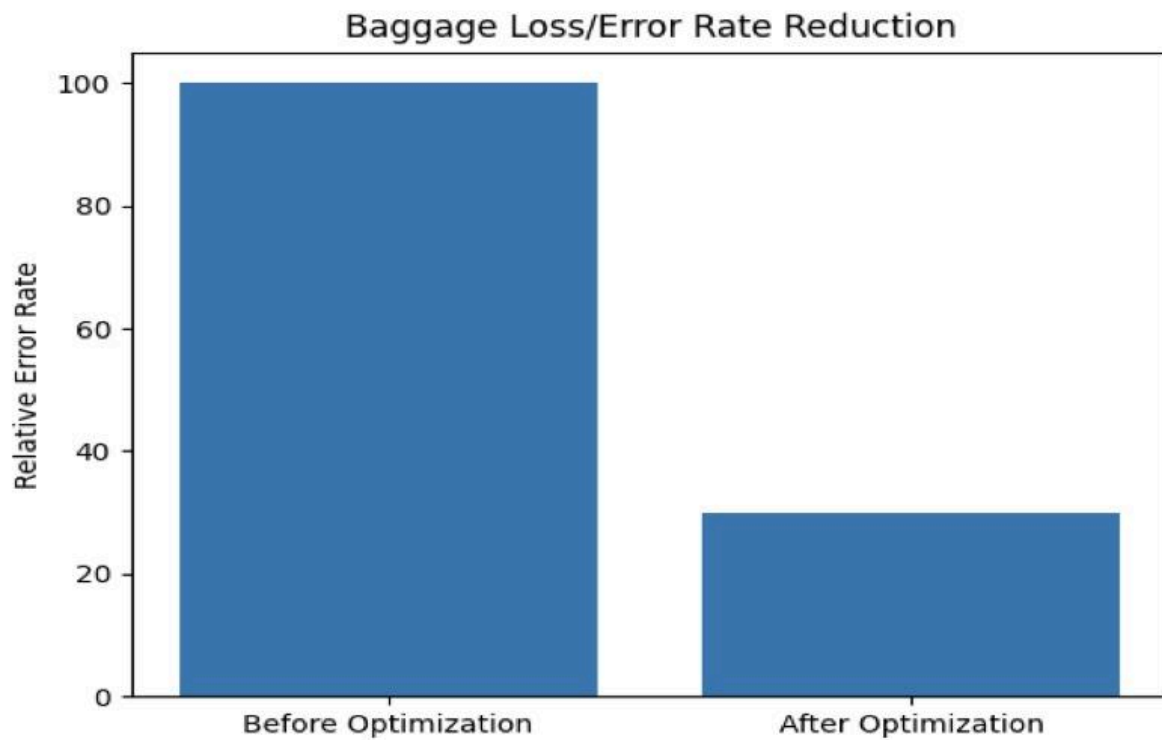
This report presents an optimized approach to aviation baggage tracking systems, integrating improved database logic, real-time tracking, and enhanced scanning technologies. Visual graphs are included to support performance improvements.



1. Problem Overview

Traditional baggage systems rely on barcode-only scanning and event-based tracking. This leads to missing scans, data mismatches, and lack of real-time visibility, which are primary causes of lost baggage.

Error Rate Reduction



The graph shows a significant reduction in baggage loss rates after implementing RFID tracking and database validation logic.

2. Optimized System Flow

[Check-in] ↓

[Barcode + RFID Tagging] ↓

[Zone-Based Tracking] ↓

[Sorting Validation] ↓

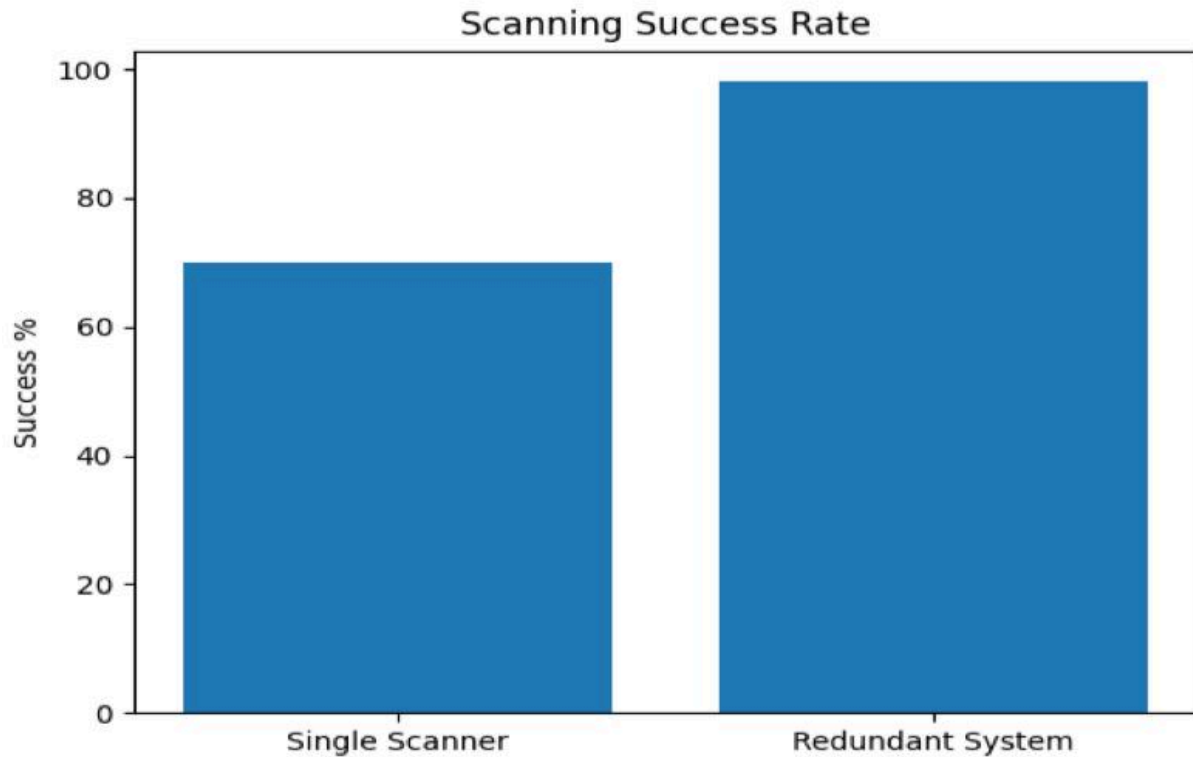
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[Aircraft]

3. Scanning Optimization

To eliminate scan failures, redundant scanning systems and RFID gates are introduced, ensuring continuous data capture.

Scanning Success Rate



The success rate increases significantly when moving from a single scanner system to a redundant multi-scanner + RFID system.

4. Smart Database Logic

IF actual_location $\neq$ expected_location AND delay > threshold $\rightarrow$ ALERT triggered $\rightarrow$ Staff notified immediately

This logic ensures that errors are detected proactively rather than after baggage is already lost.

5. Conclusion

The integration of RFID, real-time tracking, and intelligent database validation significantly enhances system reliability, reduces lost baggage incidents, and improves passenger satisfaction.